



3,400,000 stars' DNA investigated

Scientists reveal they have checked the DNA of 3,400,000 stars in the Milky Way which will assist them in finding siblings of the sun. <https://digit.in/starsdna>



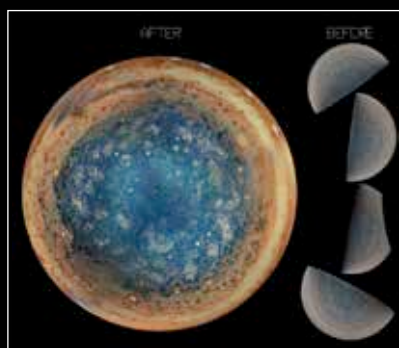
NASA to study 'marsquakes'

NASA will launch a spacecraft to Mars in May from California to study seismic motions "marsquakes" on the Red Planet. <https://digit.in/marsquakes>

the rings, while the data from 3.0 microns is used for the day side of the planet.

The Imaging Science Subsystem on board the Cassini spacecraft had two cameras, a narrow angle camera and a wide angle camera. Each camera hosted a set of rotating filters, so that the object under observation could be photographed in various wavelengths. There were eighteen such filters on the cameras. This is incredibly useful for science, NASA has made available all the raw images taken from the imaging instruments on board the Cassini spacecraft to the general public. By and large, when NASA releases the images, they try to replicate what the human eye would have seen, if it had been at the location instead of the spacecraft.

The Hubble Space Telescope has seen a series of upgrades to its imaging instruments. Currently, it uses the Wide Field Camera 3. The telescope can image objects under



An image of the south pole of Jupiter, processed by Isabela Laufer

the same object, captured from five different instruments. The Very Large array provided the radio data, which consists of wavelengths many kilometres in length. The radio data is represented by the colour red. The infrared data was captured by the Spitzer Space Telescope, and is represented by yellow. The visible light has been reduced to green, and this image was captured by the Hubble Space Telescope. The XMM-Newton provided the ultraviolet data, represented by blue. Finally, the Chandra X-ray Observatory captured the X-Ray data, represented by purple. When all of these come together, you have a truly spellbinding image capturing an immense amount of detail. All the stars appear blue, because most of the radiation they emit is in the ultraviolet wavelengths. This is not how the Crab Nebula, or the background stars, would appear to the human eye.

The JunoCam imaging instrument on board the Juno spacecraft is responsible for providing us with some of the best images of Jupiter. However, this is not strictly a scientific instrument. Although it has been used for scientific investigations, the primary purpose of the camera is to allow NASA to engage with the public, and get the image processing community involved. Towards this end, NASA rarely processes the images themselves, and instead have invited the general public to play around with the RAW images, and upload their final results on the JunoCam website.



A composite image of the Crab nebula based on data captured by five telescopes.

observation in the ultraviolet, visible and near infrared wavelengths. The RAW images captured by the Hubble Space Telescope are also in black and white. These are layered on in three channels, to produce the final images.

This image of the Crab Nebula, is actually composed of images of

The raw images are made up of strips of black and white images, captured with red, blue, green and near infrared filters. The images from the filters have to be lined up to create the final image.

The landscapes, rock formations, and even the selfies from Mars are white balanced. The image information is changed in relation to a specific pixel in the image, which is assumed to be white. This is done so that viewers can understand the formations on an alien terrain, as if they were seeing the Red Planet with lighting conditions similar to the daytime on Earth. If the humans would observe Mars for themselves, then they would see a more orangish-reddish hue, with the details appearing blurry, and the differences



Calibrated image of Mars on the left, white balanced image on the right.

between differently coloured rocks being less distinct. Even these images incorporate additional data and are processed, from special instruments that add to the images captured by the camera. In the raw data the colours are actually inaccurate, and the details are even more blurry.

Almost every image that NASA releases to the public, has a dedicated web page with a caption for the image, and the details of the post processing done, and why the image has been processed in a particular way. However, this information does not accompany the images through all the media where they are distributed, from newspapers to the internet. In the end, the viewers have little idea on what went on behind the scenes to create the stunning images. 📷